



Master Presenter's Manual

for

**Large Data Management
with MongoDB**

Curriculum Code: OV-1522

**Release Date: March, 2024
Version 1.0**

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	0

Table of Contents

S#	Details	Page No.
1.	Introduction	1
2.	Information on Module Allocation	1
3.	Week-wise Session Schedule	2
4.	Session Coverage	3
5.	Deliverables	6
6.	Library References	7
7.	Evaluation Strategy	8
8.	Classroom and Lab Pre-requisites	10

1. Introduction

This Presenter's Manual is to be used for the **Large Data Management with MongoDB** course.

After the completion of this course, students will be able to:

- ❖ Understand MongoDB concepts, features, architecture and data model, and learn to install, configure, and monitor open-source NoSQL databases.

2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on the module along with a break-up of the hours being spent on the module.

This course is based on theory and practice lab session. A session is of two hours duration.

Module Wise Session Allocation

Module	Online Hours	Skill Code
Managing large datasets with MongoDB	32	399

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

The table below shows the details of the module, the number of hours being spent on the module along with a break-up of the hours being spent on each session.

3. Week Wise Schedule

Topic \ Week		1	2	3	4
Managing large datasets with MongoDB	D1	TL1	TL5	TL9	TL13
	D2	TL2	TL6	TL10	TL14
	D3	TL3	TL7	TL11	TL15
	D4	TL4	TL8	TL12	TL16

Where,

D = Day, TL= Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	MONGODB – TL1	All the topics as listed below from Session 1 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 1 - Introduction to MongoDB</u> • Databases • An Overview of Types of DBMS • RDBMS VS NoSQL • Overview of MongoDB • What is MongoDB and how it works • Database, Collection, and Document • Getting started with MongoDB • MongoDB Server • MongoDB Cloud • MongoDB tools • Install MongoDB Shell • Install MongoDB Compass • Connecting the server with MongoDBShell	Managing Large Datasets with MONGODB SG-Session 1 XP-Session 1 TG-Session 1
2	MONGODB – TL2	All the topics as listed below from Session 2 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 2 - Working with MongoDB Databases</u> • Data Types in MongoDB • Databases and Collections • Create Database • Create Collection • Insert Document • Query Document • Update Document • Delete Document • Drop Collection • Drop Database	Managing Large Datasets with MONGODB SG-Session 2 XP-Session 2 TG-Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	MONGODB – TL3	All the topics as listed below from Session 3 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 3- MongoDB Operators</u> • Introduction to MongoDB operators • Query and Projection Operator • Comparison • Logical • Element • Array • Projection • Miscellaneous • Update Operator • Field • Array	Managing Large Datasets with MONGODB SG-Session 3 XP-Session 3 TG-Session 3
4	MONGODB – TL4	The Try It Yourself questions of Session 1 to Session 3 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session1 to 3
5	MONGODB – TL5	All the topics as listed below from Session 4 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 4 – Aggregation Pipeline</u> • Introduction to Aggregation Pipeline • Aggregation Pipeline Stages • Aggregation Expressions	Managing Large Datasets with MONGODB SG-Session 4 XP-Session 4 TG-Session 4
6	MONGODB – TL6	All the topics as listed below from Session 5 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 5 – Database Commands</u> • Introduction to Database Commands • Authentication commands • Write operation commands • Geospatial commands • Aggregation commands	Managing Large Datasets with MONGODB SG-Session 5 XP-Session 5 TG-Session 5
7	MONGODB – TL7	All the topics as listed below from Session 6 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 6 – MongoDB Shell Methods</u>	Managing Large Datasets with MONGODB

Session No.	Session Title	Session Details	Deliverables' Mapping
		• Introduction to MongoDB Shell methods • Collection methods • Database methods • User management methods • Role management methods	SG-Session 6 XP-Session 6 TG-Session 6
8	MONGODB – TL8	The Try It Yourself questions of Session 4 to Session 6 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session 4-6
9	MONGODB – TL9	All the topics as listed below from Session 7 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 7 – MongoDB Indexing</u> • Introduction to Indexes • Multi-key Index • Hashed Index • Create Index • Drop Index	Managing Large Datasets with MONGODB SG-Session 7 XP-Session 7 TG-Session 7
10	MONGODB – TL10	All the topics as listed below from Session 8 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 8 – Replication and Sharding</u> • Replication • What is replication? • Deployment Architecture • Replication Commands and Methods • Implement Replication Using an Example • Sharding • What is Sharding? • Cluster Components • Shard Keys • Sharding Commands and Methods • Implement Sharding Using an Example	Managing Large Datasets with MONGODB SG-Session 8 XP-Session 8 TG-Session 8
11	MONGODB – TL11	All the topics as listed below from Session 9 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session.	Managing Large Datasets with MONGODB

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 9 – Transaction Management</u> • Introduction to MongoDB Transactions • Transaction Properties • Transactions and Sessions • Working With Transactions	SG-Session 9 XP-Session 9 TG-Session 9
12	MONGODB – TL12	The Try It Yourself questions of Session 7 to Session 9 of <i>Managing Large Datasets with MongoDB</i> book should be covered in this session.	Managing Large Datasets with MONGODB Session 7-9
13	MONGODB – TL13	All the topics as listed below from Session 10 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 10 – MongoDB Tools</u> • Connect MongoDB Compass with MongoDB deployment • Manage Database Using Compass • Install MongoDB BI Connector • Connect MongoDB BI Connector with MongoDB deployment • Create a system Data Source Name • Manage Data Between MongoDB collection and Microsoft Excel	Managing Large Datasets with MONGODB SG-Session 10 XP-Session 10 TG-Session 10
14	MONGODB – TL14	All the topics as listed below from Session 11 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 11 – MongoDB Cloud</u> • Get Started with Atlas • Access MongoDB Atlas Cluster • Import Data into Atlas Cluster From MongoDB Instance • Export Data from Atlas Cluster Into MongoDB Instance • Administer MongoDB Cluster	Managing Large Datasets with MONGODB SG-Session 11 XP-Session 11 TG-Session 11
15	MONGODB – TL15	All the topics as listed below from Session 12 of <i>Managing Large Datasets with MONGODB</i> book should be covered in this session. <u>Session 12 – MongoDB Database Connectivity with Python</u>	Managing Large Datasets with MONGODB SG-Session 12 XP-Session 12 TG-Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		- Overview of connecting Python with MongoDB - Install PyMongo driver - Create a Database and Collection Using Python - Insert Documents Using Python - Query Documents Using Python	
16	MONGODB – TL16	The Try It Yourself questions of Session 10 to Session 12 of Managing Large Datasets with MongoDB book should be covered in this session.	Managing Large Datasets with MONGODB Session 10-12

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Students:

Learner's Guide	eBook consisting of the knowledge to be imparted (wherever applicable)
Assignments	Case study/scenario based problems to be solved by learners
Article	Additional knowledge in the form of articles that are referred from Web
Practice Questions	Additional practice questions for learners at the end of each session

Module	Book Code	Book Name
KIT	OV-1522-KIT01-01-A	Kit of 1 title for OV-1522 MongoDB
Managing Large DataSets with MongoDB	AINNMNGDB10823E000	Managing Large Databases Using MongoDB

➤ **Deliverables for Faculty**

Presenter's Manual	A training delivery plan for the trainers
Trainer's Guide	A session wise guide for the trainers
Classroom Presentations	Session wise PowerPoint presentations as teaching aid, wherever applicable

Module	Book Code	Book Name
KIT	OV-1522-KIT01-01-A	Kit of 1 title for OV-1522 MongoDB
Managing Large DataSets with MongoDB	AINNMNGDB10823E000	Managing Large Databases Using MongoDB

6. Library References

➤ MongoDB: The Definitive Guide by Kristina Chodorow and Michael Dirolf
➤ Mastering MongoDB 4.x by Wouter Thielen
➤ Data Modeling for MongoDB: Building Well-Designed and Supportable MongoDB Databases by Steve Hoberman, William Steve Hoberman

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination (Objective)
- Practical Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for PASS

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for CREDIT and

Overall Weighted Marks $\geq 75\%$ qualifies for DISTINCTION

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of the Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50

Duration: 1 hour

Type: Objective

S#	Name of the Subject	Marks
1.	Managing Large DataSets with MongoDB	50

Final Examination 2 (R1)

Marks: 50

Duration: 2 hours

Type: Practical

S#	Name of the Subject	Marks
1.	Managing Large DataSets with MongoDB	50

Evaluation Details

OES-50%

Practical-50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the sessions have been described as follows:

8.1 Software Requirements

Microsoft Windows 11
MongoDB 6.0

8.2 Recommended Hardware

Intel Core i5/i7 Processor or higher
8 GB or 16 GB RAM
1 TB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

~~ End of Document ~~